CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter reviews related literature and studies relevant to analyzing medicine procurement and distribution data in Philippine public health facilities using unsupervised clustering techniques. The reviewed works establish the context of medicine availability challenges, including disparities in access, inefficiencies in supply allocation, and systemic risks such as shortages and uneven distribution. In addition, this chapter synthesizes research on public health resource management and health data analytics to explain how governance, budgeting, logistics systems, and data-driven monitoring influence medicine availability outcomes. Finally, the chapter examines methodological foundations and applied studies in unsupervised machine learning and clustering, highlighting how clustering approaches can uncover hidden patterns in large, heterogeneous, and unlabeled datasets. Collectively, these studies justify the use of clustering analysis to identify distribution patterns, detect anomalies, and support evidence-based decision-making for improving medicine supply allocation and availability in public health systems.

Medicine Distribution and Availability Studies

Existing literature indicates that medicine availability in public health systems remains uneven and fragile despite policy reforms and prioritization initiatives. Large-scale studies and systematic reviews report that essential

medicines are generally more available than non-essential medicines, particularly in public sector facilities and in low- and middle-income countries. However, overall availability levels remain inadequate, with persistent shortages and substantial disparities observed across regions, income groups, facility types, and therapeutic categories (Bazargani et al., 2015; Zhang et al., 2023). This evidence suggests that essential medicines policies influence procurement priorities but do not ensure consistent or equitable access across the health system.

Regional and geographic disparities consistently emerge in studies of medicine distribution. Longitudinal and cross-sectional analyses from China show that the availability of essential medicines has improved only marginally over time, while western and rural regions continue to exhibit significantly lower availability compared to eastern and central regions (Zhang et al., 2023). Similar patterns are reported in decentralized health systems and geographically dispersed settings, where facilities serving remote or sparsely populated areas experience recurrent stockouts and irregular delivery schedules. In many cases, the absence of standardized and continuous monitoring data limits the ability of authorities to assess distribution performance and respond promptly to emerging shortages.

Policy-driven reforms in medicine procurement have also produced mixed outcomes for distribution performance. Studies examining centralized purchasing and restrictions on the number of approved distributors indicate that cost-saving measures can unintentionally weaken medicine delivery when geographic

dispersion and demand variability are insufficiently considered. Evidence from primary care institutions in China shows that reducing the number of distributors led to declines in delivery and receipt rates, particularly in rural clinics, thereby widening existing access gaps (Yang, Huang, & Liu, 2017). These findings demonstrate that efficiency gains at the procurement stage do not necessarily translate into improved distribution outcomes and may introduce new system-level risks when logistical constraints are overlooked.

Beyond formal procurement policies, broader structural pressures within health systems play a critical role in shaping medicine availability. Research examining health systems under conditions of globalization, fiscal constraints, regulatory complexity, and public health emergencies shows that these combined pressures weaken supply security and increase the frequency of medicine shortages (Jenzer, Groesser, & Miljković, 2023). Even in relatively well-resourced settings, hospitals and pharmacies often rely on emergency measures such as importation or alternative production pathways to maintain medicine access. These practices highlight the fragility of medicine supply chains and underscore the need for continuous monitoring and proactive management.

In the Philippine context, medicine distribution challenges are closely linked to financial constraints and decentralized local practices. Studies of barangay health centers demonstrate that limited budgets and inefficient allocation mechanisms leave primary care facilities vulnerable to supply shortages, particularly when each facility must independently manage procurement and inventory (Bucu et al., 2023). Community-level evidence from

the Botika ng Barangay program further illustrates how informal financial practices influence medicine availability. Qualitative interviews reveal that unpaid medicine credit arrangements undermine the sustainability of local drug outlets, preventing timely replenishment even when initial supply levels appear sufficient (Collado et al., 2021). These findings indicate that medicine availability depends not only on procurement volumes but also on financial governance, local practices, and institutional coordination.

Collectively, these studies demonstrate that medicine availability challenges are systemic, multidimensional, and deeply embedded within public health supply structures. Persistent shortages, regional inequities, and unintended consequences of procurement reforms are observable through aggregated procurement and distribution data. However, much of the existing literature relies on descriptive analysis, policy evaluation, or qualitative inquiry. This body of evidence underscores the need for data-driven analytical approaches capable of uncovering latent distribution patterns, identifying facility-level and regional imbalances, and detecting systemic risks within large-scale, unlabeled medicine procurement and distribution datasets.

Healthcare Resource Management in Public Health Systems

Research on healthcare resource management consistently emphasizes that medicine availability is closely tied to broader financial, logistical, and governance structures within public health systems. Studies focusing on financial management highlight that stable funding mechanisms, transparent budgeting

processes, and efficient resource allocation are strongly associated with improved service quality and access, particularly in low- and middle-income countries (Abidin & Tobing, 2025). Conversely, weak financial governance often results in inequitable resource distribution and service gaps, limiting the capacity of public health facilities to maintain reliable medicine supplies. Although such studies do not always focus explicitly on medicines, they provide critical system-level context for understanding how procurement and distribution inefficiencies are frequently rooted in budgetary constraints and allocation practices.

Logistics management literature further demonstrates that medicine availability is an outcome of interconnected operational processes rather than procurement volume alone. Reviews of public health logistics systems identify demand estimation, procurement planning, inventory control, storage, and distribution as core functions that directly influence stock stability (Priyanga & Rajalakshmi, 2024). Weaknesses in these processes commonly lead to stockouts, overstocking, and medicine expiry. Established inventory classification techniques, such as ABC and VED analyses, are frequently recommended to support rational prioritization under limited resources. The increasing use of real-time information systems to improve stock visibility and decision-making further underscores the importance of data-driven approaches in managing medicine supply chains. Together, these findings frame medicine availability as a function of systematic logistics management rather than isolated procurement decisions.

Empirical studies applying advanced analytics to healthcare resource management reinforce this perspective. Research on machine learning-based inventory management and demand forecasting in public healthcare centers identifies poor forecasting accuracy, limited inventory visibility, and suboptimal sourcing as major contributors to medicine stockouts and overstocking (Bhat et al., 2024). By demonstrating that medicine demand and inventory behavior follow identifiable patterns within aggregated procurement and utilization data, these studies provide evidence that data-driven analysis can support more efficient allocation and cost optimization. Although such work often relies on supervised or predictive models, it complements unsupervised approaches by confirming that systemic risks are embedded within routinely collected supply data.

Beyond financial and logistical processes, governance and human resource management also shape medicine availability through indirect but significant pathways. Studies examining public healthcare systems during periods of crisis, such as the COVID-19 pandemic, show that structural reforms aimed at cost containment can expose vulnerabilities when systems face sudden demand shocks (Aksenova et al., 2020). Shortages of personnel, hospital capacity, and protective equipment often coincide with medicine shortages, limiting system responsiveness. Similarly, research on public health services in India demonstrates that informal governance practices and non-transparent management structures undermine accountability and equitable resource distribution, ultimately weakening service delivery reliability (La Forgia et al.,

2015). These findings suggest that medicine availability cannot be separated from broader institutional performance and governance quality.

Evidence from high-resource healthcare settings further illustrates the interdependence of resource planning and medicine supply. Studies of hospital-level resource management during the COVID-19 pandemic highlight how proactive, data-informed strategies, including capacity reallocation and workforce coordination, can mitigate shortages and improve system resilience (Lee et al., 2020). Although such research focuses primarily on critical care operations, it demonstrates that integrated resource management approaches contribute to maintaining access to essential medical supplies during periods of extreme strain.

Collectively, this body of literature demonstrates that medicine procurement and distribution challenges are embedded within wider public health resource management systems. Financial governance, logistics processes, human resource management, and institutional decision-making jointly influence the stability of medicine supply. These studies also indicate that inefficiencies and risks are reflected in operational data generated by public health systems. This evidence supports the application of data-driven analytical approaches, including unsupervised clustering, to aggregated procurement and distribution records in order to identify systemic imbalances, inefficiencies, and risks affecting medicine availability across public health facilities.

Health Data Analytics in Public Health Decision-Making

Health data analytics is increasingly recognized as a critical enabler of evidence-based decision-making in public health systems. Literature on analytics platforms demonstrates that structured data pipelines, feature engineering, and real-time dashboards allow public health managers to monitor system performance, forecast trends, and identify operational risks such as stockouts and service bottlenecks (Nwokocha et al., 2025). The use of analytics for spatial aggregation, key performance indicator tracking, and risk identification illustrates how large-scale, routinely collected health system data can be transformed into actionable insights that support both policy formulation and operational planning.

The expansion of data-driven decision-making has been further supported by advances in analytics infrastructure, particularly cloud-based systems. Conceptual and applied studies emphasize that the growing volume and complexity of digitized health data require integrated analytics environments capable of supporting the full data lifecycle, from acquisition and preprocessing to model deployment and continuous monitoring (Patel, 2024). Embedding analytics into routine public health operations enables more timely responses to emerging risks and supports strategic planning related to resource allocation and system performance. Although such work often focuses on infrastructure rather than specific analytical techniques, it provides important context for understanding how large-scale public health datasets can be systematically analyzed.

Broader reviews of healthcare analytics reinforce the role of data-driven approaches in improving resource management and operational efficiency. Syntheses of analytics applications across healthcare domains show that integrating data from electronic health records, operational systems, and other digital sources supports informed planning, optimized resource utilization, and strategic decision-making in complex health systems (Arowoogun et al., 2024). While these studies are not limited to medicine procurement or distribution, their emphasis on analytics-driven resource allocation directly informs public health supply system analysis.

The credibility and usefulness of analytics outputs depend strongly on data quality, transparency, and methodological rigor. Research on real-world data analytics highlights the value of rapid-cycle analysis of routinely collected healthcare data in generating timely and policy-relevant evidence (Schneeweiss & Glynn, 2018). Frameworks such as MVET emphasize that analytics must produce evidence that is meaningful, valid, expedited, and transparent in order to inform decision-making. These principles are directly applicable to public health supply analytics, where procurement and distribution data must be analyzed in ways that support trustworthy monitoring and policy intervention.

At the same time, studies examining stakeholder perspectives reveal persistent challenges in translating analytics potential into operational use. Research within health systems shows that although big data analytics is widely acknowledged as valuable for system-level planning, its routine application is often constrained by concerns related to data quality, system integration,

governance, and organizational readiness (Weerasinghe et al., 2022). These findings highlight the importance of institutional acceptance, stakeholder engagement, and trust in data for analytics to meaningfully influence decision-making processes.

Governance-focused literature further emphasizes that effective public health analytics requires appropriate institutional frameworks. Research on data accessibility and governance models demonstrates that integrated analytics infrastructures, supported by ethical data-sharing mechanisms and transparent governance, enhance the capacity of public health systems to evaluate policies, monitor risks, and respond to emerging challenges (Mählmann et al., 2017). Framing health data as a public good reinforces the need for coordinated and responsible data use in public sector decision-making.

Collectively, this body of literature demonstrates that health data analytics plays a foundational role in transforming large-scale, aggregated, and non-personal health system data into actionable knowledge. While much of the existing work focuses on predictive analytics, visualization, and governance, it establishes the conditions under which advanced analytical methods can support public health decision-making. These insights provide a strong rationale for applying data-driven approaches, including unsupervised clustering, to medicine procurement and distribution data in order to enhance monitoring, identify systemic risks, and support evidence-based resource allocation in public health systems.

Unsupervised Machine Learning in Health Data Analysis

Unsupervised machine learning has become an important analytical approach for exploring complex and high-dimensional healthcare datasets, particularly in contexts where predefined labels or outcome variables are unavailable. Methodological overviews emphasize that techniques such as k-means clustering, hierarchical clustering, principal component analysis, factor analysis, and density-based methods are well suited for identifying latent structures and hidden patterns in heterogeneous health data (Eckhardt et al., 2023; Wong, 2021). Conceptual and methodological literature further establishes clustering as a dominant class of unsupervised learning techniques across data science applications, with systematic reviews identifying k-means clustering, hierarchical clustering, and dimensionality reduction methods as the most frequently applied unsupervised approaches in large-scale, unlabeled datasets (Alloghani et al., 2020). These approaches are especially valuable in healthcare settings characterized by data complexity, variability across units, and incomplete labeling, where exploratory analysis is required to uncover meaningful groupings and relationships, providing a strong methodological foundation for applying clustering techniques to health system data where data heterogeneity and high dimensionality are common.

Empirical studies in healthcare demonstrate the practical utility of unsupervised learning for pattern discovery and subgroup identification. Research applying clustering techniques to electronic health records has shown that unsupervised methods can reveal clinically meaningful subpopulations without reliance on predefined categories. Studies using k-means and related clustering approaches have identified patient groups with distinct laboratory profiles, treatment patterns, and outcome risks, illustrating how heterogeneous health data can be organized into interpretable structures that support clinical understanding and resource planning (Hyun et al., 2020). More advanced unsupervised models further highlight methodological considerations relevant to system-level analysis. Probabilistic approaches applied to electronic health record data demonstrate that unsupervised learning can uncover latent disease clusters and subgroup structures while also revealing potential biases introduced by demographic variables (Wang et al., 2020). Comparisons between models show that clusters may reflect demographic drivers rather than underlying mechanisms if such factors are not adequately accounted for, emphasizing the importance of careful feature selection, validation, and interpretation when using unsupervised clustering for risk identification and pattern discovery in health-related datasets.

Methodological rigor in evaluating clustering outcomes is also emphasized in studies comparing multiple unsupervised techniques. Research examining the identification of clinical subtypes through clustering highlights the need to assess cluster stability, structural validity, and interpretability rather than relying solely on algorithmic output (Alexander et al., 2021). Collectively, this body of literature establishes unsupervised machine learning as a robust and appropriate approach for analyzing complex, unlabeled healthcare data. Clustering techniques enable the discovery of latent patterns, support system-level understanding, and provide a data-driven basis for identifying risk profiles in environments where predefined performance labels are absent. These methodological insights directly support the application of unsupervised clustering to aggregated, non-personal medicine procurement and distribution data, where uncovering hidden supply patterns and systemic risks is central to improving public health decision-making.

Recent applications of unsupervised clustering in healthcare procurement further demonstrate its relevance for decision support in complex purchasing environments. Wang (2023) examined the use of k-means clustering within an intelligent hospital procurement supplier management system, proposing an optimized variant that integrates particle swarm optimization to improve clustering stability and evaluation performance. The study showed that clustering can effectively group suppliers based on multidimensional procurement attributes, supporting supplier evaluation, allocation strategies, and risk-informed decision-making in hospital supply chains. While the research emphasizes algorithmic optimization rather than exploratory pattern discovery, it provides empirical evidence that unsupervised clustering methods are applicable and valuable in healthcare procurement contexts, supporting the use of clustering as a decision-support mechanism in public health supply systems where grouping entities based on observed procurement and supply characteristics can inform more structured and evidence-based management decisions.

Clustering Techniques in Public Health and Medicine Distribution

Clustering techniques have been increasingly applied to public health and healthcare-related data to analyze distribution patterns, identify spatial disparities, and support evidence-based planning. Studies integrating clustering with multi-source and spatial data demonstrate that such approaches are effective for examining regional variation and structural concentration within healthcare systems. Research combining principal component analysis, hotspot

analysis, and spatial clustering shows that a limited set of underlying factors often explains a substantial proportion of variance in healthcare-related distribution patterns, revealing distinct geographic groupings characterized by uneven development and resource concentration (Wang et al., 2025). Spatial clustering has also been widely used as a decision-support tool in public health planning, particularly when integrated with geographic information systems and data visualization, enabling large-scale health datasets to be transformed into interpretable spatial and temporal patterns that guide targeted interventions (Rahman, 2022). Although such studies often focus on disease surveillance, the underlying methodological framework is directly applicable to medicine distribution analysis, where spatial clustering can be used to identify geographic imbalances in medicine availability, regional demand variation, and areas at heightened risk of supply disruption, particularly in decentralized public health systems.

Methodological advances in clustering algorithms further expand their applicability to complex health system and distribution data. Research proposing novel unsupervised clustering approaches demonstrates that high-dimensional data can be transformed into lower-dimensional representations that preserve intrinsic structure while reducing sensitivity to noise and outliers (Yang et al., 2018). The ability to identify clusters with arbitrary shapes without requiring prior specification of cluster numbers is particularly relevant to medicine procurement and distribution datasets, which are often heterogeneous, high dimensional, and unlabeled. Related methodological contributions address limitations of traditional

clustering techniques when applied to real-world medical and health system data. Overlapping clustering approaches allow data instances to belong to multiple clusters, reflecting the non-exclusive and intersecting characteristics commonly observed across facilities, regions, and therapeutic categories (Khanmohammadi et al., 2017). Density- and structure-aware clustering methods further enhance the ability to analyze complex distribution patterns, with algorithms incorporating local density estimation and path-based distance measures shown to outperform conventional distance-based methods in detecting arbitrary-shaped and manifold clusters (Chen et al., 2022). These capabilities are particularly important for public sector medicine distribution data, where patterns are shaped by geographic dispersion, facility heterogeneity, and temporal demand fluctuations, supporting more nuanced identification of supply anomalies and systemic vulnerabilities.

Empirical applications of clustering to medicine inventory and supply data provide practical evidence of its value for supply management. Studies applying k-means clustering to hospital drug inventory records demonstrate that combining clustering with inventory classification techniques can effectively group medicines based on expenditure, variability, movement, and clinical importance (Pintakiew & Laosiritaworn, 2024). The resulting clusters support differentiated inventory management strategies and contribute to improved allocation efficiency and reduced risk of shortages. Although such applications are often situated in private healthcare settings, they illustrate the practical relevance of clustering techniques for organizing medicine data and informing

supply decisions. Collectively, this literature demonstrates that clustering techniques offer flexible and powerful tools for analyzing medicine distribution and healthcare-related supply data. By accommodating spatial variation, high dimensionality, overlapping characteristics, and complex data structures, clustering supports the identification of distribution imbalances, demand patterns, and systemic risks, providing a strong foundation for applying unsupervised clustering to aggregated medicine procurement and distribution data in public health systems.

Methodological evidence from supply chain and logistics analytics further supports the use of combined dimensionality reduction and clustering techniques for inventory and distribution analysis. Gonçalves, Cortez, and Carvalho (2021) applied principal component analysis (PCA) in conjunction with k-means clustering to analyze multi-item, multi-supplier inventory data, demonstrating that descriptive, unsupervised data mining can uncover latent patterns and risk profiles embedded in aggregated inventory data. Comparative studies further demonstrate the value of applying multiple unsupervised clustering paradigms to large-scale operational datasets. León Villalba and González La Rotta (2020) compared density-based (DBSCAN) and centroid-based (k-means) clustering methods using large logistics delivery records, highlighting how DBSCAN effectively captures dense geographic groupings and irregular spatial structures, while k-means provides more compact and interpretable partitions. Additional evidence from logistics analytics reinforces the effectiveness of density-based clustering for identifying high-concentration distribution patterns, with DBSCAN

shown to handle noise and outliers while structuring large and heterogeneous spatial datasets (Li et al., 2020). Although situated outside the healthcare sector, these studies are methodologically relevant to public health medicine distribution, where procurement and distribution data similarly exhibit high dimensionality, spatial dispersion, and heterogeneous density patterns. Together, they provide strong justification for integrating dimensionality reduction and complementary clustering techniques to explore structural properties and identify systemic risks within public health supply systems.

Application of Clustering in Healthcare Resource Management

Clustering techniques have been widely applied in healthcare resource management to support planning, demand analysis, and system-level decision-making in environments characterized by complex and unlabeled data. Applications of unsupervised clustering in healthcare management demonstrate that grouping data based on observed patterns can reveal trends and risks that are not readily detectable through traditional descriptive analysis. Studies integrating clustering within smart healthcare systems show that unsupervised learning can identify temporal and structural patterns in health data, supporting improved resource utilization and planning across both chronic and acute care contexts (Yang et al., 2023). Methodological reviews further emphasize clustering as a foundational analytical tool for healthcare resource management, highlighting its effectiveness in extracting meaningful patterns from large-scale, heterogeneous, and unlabeled datasets, particularly when combined with

complementary analytical techniques (Oyewole & Thopil, 2023). In healthcare settings, clustering is increasingly used to support resource optimization, system-level analysis, and sustainability planning rather than as a standalone exploratory method. These reviews also identify recurring methodological challenges, including algorithm selection, determination of optimal cluster numbers, and validation of clustering results as data scale increases, all of which are directly relevant to analyzing aggregated medicine procurement and distribution data.

Applications of clustering in healthcare analytics further demonstrate its utility when labeled data are incomplete or unreliable. Research examining clustering in disease profiling and health data segmentation shows that unsupervised methods can uncover latent structures without reliance on standardized diagnostic codes, which are often missing or inconsistent in real-world datasets (Negi & Chawla, 2021). Clustering has also been applied to healthcare utilization data to support resource allocation and service planning, with studies using utilization-based clustering of linked care datasets demonstrating that data-driven segmentation can identify distinct groups with differing patterns of service use, morbidity, and risk factors, offering more actionable insights than expert-defined classifications (Nnoaham & Cann, 2020). Although focused on population health management, these studies highlight the relevance of clustering for understanding demand patterns that directly influence medicine distribution and supply requirements. By grouping entities based solely on observed attributes, clustering supports the identification of structural patterns that can inform more effective resource management and planning in public health contexts.

Operational applications further illustrate how clustering can support managerial decision-making in complex healthcare processes. Research applying robust clustering techniques to healthcare event logs shows that grouping high-variability process data can reveal dominant behavioral patterns and improve the interpretability of operational workflows (Delias et al., 2015). While such studies focus on clinical pathways, their methodological approach is applicable to medicine procurement and distribution processes, which similarly generate complex and variable transaction data. Clustering aggregated procurement and distribution records can therefore support the identification of dominant supply patterns, inefficiencies, and potential systemic risks within public health facilities. Collectively, this body of literature demonstrates that clustering techniques provide a practical and flexible analytical framework for healthcare resource management. By enabling the analysis of complex, heterogeneous, and unlabeled data, clustering supports demand analysis, resource optimization, and system-level insight generation, reinforcing the suitability of unsupervised clustering for analyzing aggregated, non-personal medicine procurement and distribution data.

Evidence from manufacturing and industrial procurement contexts further illustrates the role of clustering as a decision-support tool in complex purchasing environments. Kujawińska et al. (2018) applied both hierarchical clustering (Ward's method) and generalized k-means clustering to support purchasing decisions for manufacturing materials in an Industry 4.0 setting. Using a dataset of over 200 material combinations, the study demonstrated that unsupervised clustering can effectively organize high-dimensional procurement data into meaningful groups that inform material selection decisions. Although conducted within a manufacturing domain, the study reinforces the broader applicability of clustering-based data mining for procurement decision-making under conditions of data complexity and uncertainty. This methodological perspective is directly relevant to public health supply systems, where medicine procurement and distribution data similarly require exploratory analysis to transform large volumes of operational data into actionable insights.

Synthesis of Related Literature and Studies

The reviewed literature collectively demonstrates that challenges in medicine availability and distribution within public health systems are systemic rather than isolated, arising from interrelated structural, managerial, and data-related factors. Studies focusing on medicine distribution and availability consistently report persistent shortages, regional disparities, and inefficiencies in public sector supply chains despite policy interventions such as essential medicines lists and centralized procurement systems. Large-scale quantitative

analyses and systematic reviews show that aggregated availability data can reveal inequities across regions, facility types, and therapeutic categories, while qualitative and localized studies highlight how community-level practices and socio-economic dynamics further shape access to medicines. Together, these findings underscore the importance of continuous and data-driven monitoring approaches capable of identifying latent distribution patterns and systemic risks.

Literature on healthcare resource management situates medicine availability within broader public health system constraints, including financial governance, logistics management, and human resource allocation. Weak budgeting processes, fragmented governance arrangements, and inefficient logistics operations have been shown to indirectly exacerbate medicine shortages and uneven distribution. In contrast, studies emphasize that data-informed planning, inventory classification methods, and proactive resource management strategies can improve system resilience and mitigate supply disruptions, particularly during public health emergencies. These works indicate that medicine procurement and distribution performance cannot be fully understood in isolation but must be examined within the wider context of institutional decision-making and resource management structures.

Research on health data analytics in public health decision-making highlights the expanding role of analytics platforms, big data infrastructures, and real-world evidence in supporting policy formulation, performance monitoring, and operational planning. This body of literature demonstrates how structured data pipelines, dashboards, and analytical frameworks translate large volumes of

routine health system data into actionable insights. At the same time, it identifies persistent challenges related to data quality, governance, and organizational readiness, suggesting that analytical outputs are effective only when supported by robust data management practices and institutional acceptance. These insights reinforce the relevance of applying advanced analytical methods to aggregated and non-personal health system data, including medicine procurement and distribution records.

Methodological research on unsupervised machine learning and clustering provides the analytical foundation for extracting meaningful patterns from complex, high-dimensional, and unlabeled healthcare datasets. Prior studies demonstrate that clustering techniques are effective for revealing latent structures, organizing heterogeneous entities, and supporting system-level analysis without reliance on predefined labels. Applications of clustering in healthcare resource management and medicine-related contexts further illustrate how these methods can identify demand patterns, operational inefficiencies, and regional disparities, thereby supporting more evidence-based resource allocation and planning. In particular, both centroid-based clustering and density-based clustering are consistently highlighted as useful approaches for uncovering structure in large-scale, unlabeled operational datasets, supporting the use of K-means clustering and DBSCAN for analyzing aggregated procurement and distribution patterns.

Taken together, the reviewed literature reveals a clear gap between the growing availability of large-scale medicine procurement and distribution data

and the systematic application of unsupervised analytical techniques to interpret such data in public health settings. While existing studies confirm the presence of distribution inefficiencies and resource constraints, few integrate clustering-based approaches to holistically analyze medicine supply patterns across public health facilities. This synthesis therefore supports the present study's focus on applying unsupervised clustering analysis using K-means and DBSCAN to aggregated medicine procurement and distribution data in order to identify distribution patterns, detect systemic risks, and inform data-driven decision-making aimed at improving medicine availability within public health systems.

Gap Bridged by the Study

The reviewed literature demonstrates substantial scholarly attention to medicine availability, healthcare resource management, health data analytics, and the application of clustering techniques in healthcare contexts. Despite these contributions, several critical gaps remain in the systematic analysis of medicine procurement and distribution within public health systems.

First, many existing studies identify medicine shortages, regional disparities, and inefficiencies in public health supply systems through descriptive statistics, policy evaluations, or qualitative methods. While these approaches are effective in documenting observed problems, they offer limited insight into the underlying distribution patterns and systemic risks that emerge from large-scale procurement and distribution data. Few studies employ unsupervised analytical techniques capable of uncovering latent structures within aggregated medicine

supply datasets, including the combined use of centroid-based and density-based clustering approaches such as K-means and DBSCAN.

Second, research on healthcare resource management emphasizes the influence of financial governance, logistics systems, and institutional capacity on medicine availability. However, these studies rarely integrate advanced data-driven analytical methods to examine procurement and distribution records in a holistic manner. As a result, overlapping patterns, multidimensional inefficiencies, and early indicators of supply disruption are often insufficiently explored or remain undetected.

Third, studies on health data analytics highlight the growing use of dashboards, predictive models, and real-world data to support decision-making in public health. Nevertheless, such approaches commonly depend on predefined indicators or labeled performance outcomes. In public health medicine supply systems, where explicit labels for risk, efficiency, or performance are frequently unavailable or inconsistent, there is limited empirical work demonstrating how unsupervised learning methods can systematically analyze and segment facilities or medicines based solely on observed operational data using K-means and DBSCAN.

Finally, although clustering techniques are widely applied in clinical research, disease surveillance, and hospital management, their application to public sector medicine procurement and distribution remains limited, particularly in developing country settings. Existing studies rarely connect methodological advances in unsupervised clustering with practical public health supply chain

analysis aimed at operational decision-making and policy support, particularly through the combined application of K-means and DBSCAN to capture both partition-based groupings and density-based structures in distribution data.

This study addresses these gaps by applying unsupervised clustering analysis using K-means and DBSCAN to aggregated, non-personal medicine procurement and distribution data from public health facilities. By moving beyond descriptive and rule-based approaches, the study identifies latent distribution patterns, systemic risks, and facility-level imbalances without reliance on predefined classifications. The research integrates perspectives from medicine availability research, healthcare resource management, and health data analytics into a unified analytical framework. In doing so, it contributes empirical evidence on how clustering-based methods can support data-driven monitoring, enhance decision-making, and improve medicine availability within public health systems.

Conceptual Framework

This study is anchored on the Input–Process–Output (I-P-O) framework to explain how medicine procurement and distribution data are transformed into meaningful insights for public health supply management. The framework assumes that medicine availability and supply performance are influenced by observable patterns within procurement and distribution activities.

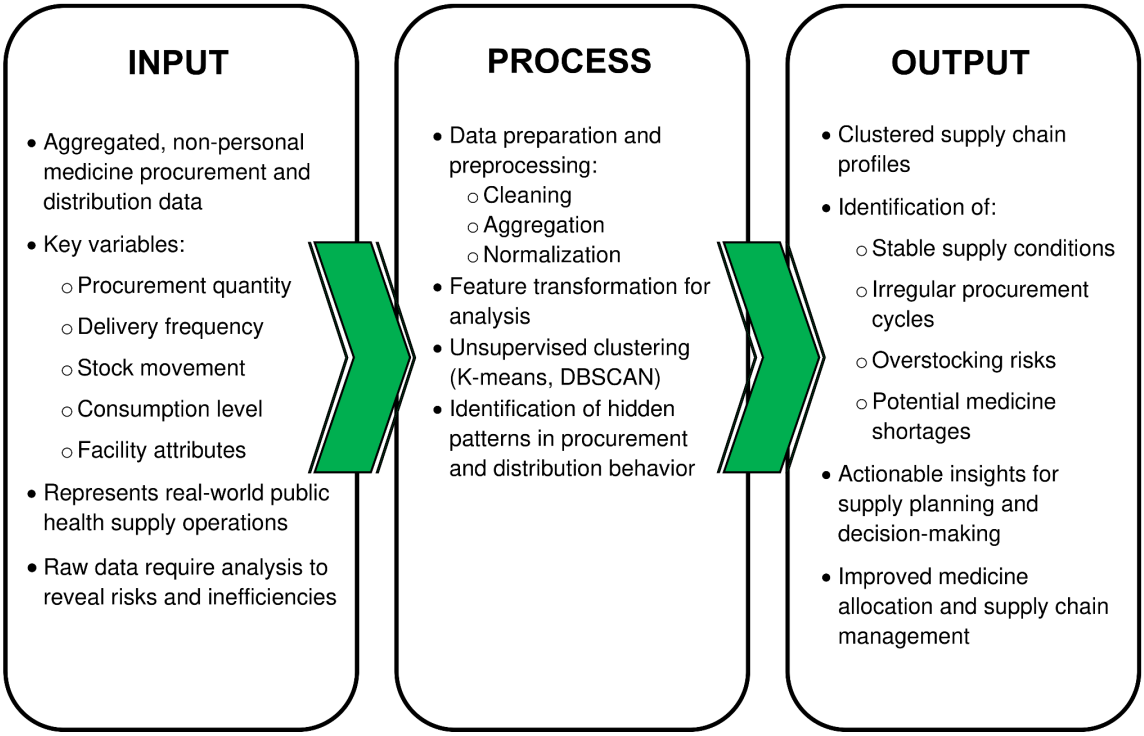

Figure 1. ***Conceptual Framework***

At the input stage, the study uses aggregated and non-personal data from public health facilities, including procurement quantities, delivery frequency, stock movement, consumption levels, and facility attributes. These inputs represent actual supply chain operations but do not directly indicate inefficiencies or risks.

The process stage involves systematic data preprocessing, such as cleaning, aggregation, and normalization, followed by the application of unsupervised clustering techniques. These analytical methods identify natural groupings in the data and reveal hidden patterns in procurement and distribution behavior without relying on predefined labels.

At the output stage, the generated clusters describe distinct medicine supply conditions, including stable supply patterns, irregular procurement cycles, overstocking tendencies, and potential risks of shortages. These outputs provide evidence-based insights that support monitoring, planning, and decision-making.